Supplementary Material: Leverage-Driven Software Architecture: A Probabilistic Framework for Architectural Decision-Making

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1 Full Lean Handle Ledger

This supplement provides the complete Lean handle ledger cited by the manuscript. It includes every handle identifier, declaration name, and source module path.

ID	Lean Handle / Source	ID	Lean Handle / Source
L9	ACS1 paper1/HandleAliases.lean	L10	ACS2 paper1/HandleAliases.lean
L11	ACS3 paper1/HandleAliases.lean	L12	ACS4 paper1/HandleAliases.lean
L13	ACS5 paper1/HandleAliases.lean	L14	ACS6 paper1/HandleAliases.lean
L15	ACS7 paper1/HandleAliases.lean	L16	ACS8 paper1/HandleAliases.lean
L17	ACS9 paper1/HandleAliases.lean	L18	ADV1 paper1/HandleAliases.lean
L19	ADV2 paper2/Ssot/HandleAliases.lean	L20	ADV3 paper2/Ssot/HandleAliases.lean
L21	ADV4 paper2/Ssot/HandleAliases.lean	L22	ADV5 paper2/Ssot/HandleAliases.lean
L23	AFM1 paper2/Ssot/HandleAliases.lean	L24	AFM10 paper2/Ssot/HandleAliases.lean
L25	AFM11 paper2/Ssot/HandleAliases.lean	L26	AFM12 paper2/Ssot/HandleAliases.lean
L27	AFM13 paper2/Ssot/HandleAliases.lean	L28	AFM14 paper2/Ssot/HandleAliases.lean
L29	AFM15 paper2/Ssot/HandleAliases.lean	L30	AFM16 paper2/Ssot/HandleAliases.lean
L31	AFM17 paper2/Ssot/HandleAliases.lean	L32	AFM18 paper2/Ssot/HandleAliases.lean
L33	AFM19 paper2/Ssot/HandleAliases.lean	L34	AFM2 paper2/Ssot/HandleAliases.lean
L37	AFM5 paper2/Ssot/HandleAliases.lean	L38	AFM6 paper2/Ssot/HandleAliases.lean
L39	AFM7 paper2/Ssot/HandleAliases.lean	L40	AFM8 paper2/Ssot/HandleAliases.lean
L41	AFM9 paper2/Ssot/HandleAliases.lean	L42	ALT1 paper1/HandleAliases.lean

2 Claim-to-Lean Handle Mapping

This section maps each paper claim to its corresponding Lean formalization.

Paper claim	**Lean handle**
Corollary 3.2: DOF = Independent Error Sources	L15, L23
Corollary 3.6: DOF-Error Monotonicity	L33
Corollary 3.13: DOF Ratio Predicts Error Ratio	L21, L42
Corollary 3.5: Linear Approximation	L16, L39
Definition 2.4: Architecture	L17, L19
Theorem 3.10: Ordering Equivalence (Exact)	L32, L39
Theorem 4.4: Leverage Composition	L18, L19
Theorem 3.9: DOF-Reliability Isomorphism	L22, L27
Theorem 3.3: Error Compounding	L20, L41
Theorem 3.1: Error Independence	L23
Theorem 3.4: Error Probability	L24, L40
Theorem 3.7: Expected Error Bound	L25, L26
Theorem 4.2: Leverage-Error Tradeoff	L29
Theorem 3.11: Leverage Gap	L30
Theorem 4.1: Leverage Maximization Principle	L28, L31
Definition 2.12: Capability Set	L37
Corollary 5.7: Unbounded Advantage	L12, L13
Theorem 4.3: Optimal Architecture	L34, L38
Definition 5.3: SSOT Architecture	L9, L11
Theorem 3.12: Testable Prediction	L30, L42

Auto summary: mapped 20/20 (full=20, derived=0, unmapped=0).

3 Proof Hardness Index

Paper handle	**Hardness profile**	**Regime tags**	**Lean support**
cor:dof-errors	unspecified	-	L15, L23
cor:dof-monotone	unspecified	-	L33
cor:dof-ratio	unspecified	-	L21, L42
cor:linear-approx	unspecified	-	L16, L39
prop:dof-additive	unspecified	-	L17, L19
thm:approx-bound	unspecified	-	L32, L39
thm:composition	unspecified	-	L18, L19
thm:dof-reliability	unspecified	-	L22, L27
thm:error-compound	unspecified	-	L20, L41
thm: error-independence	unspecified	-	L23
thm:error-prob	unspecified	-	L24, L40
thm:expected-errors	unspecified	-	L25, L26
thm:leverage-error	unspecified	-	L29
thm:leverage-gap	unspecified	-	L30
thm:leverage-max	unspecified	-	L28, L31
thm:mod-bound	unspecified	-	L37
thm:nominal-leverage	unspecified	-	L12, L13
thm:optimal	unspecified	-	L34, L38

thm:ssot-leverage	unspecified	-	L9, L11
thm: testable-prediction	unspecified	-	L30, L42

Auto summary: indexed 20 claims by hardness profile (unspecified=20).

4 Assumption Ledger

4.1 Assumption Ledger (Auto)

Source files.

- `Leverage/Examples.lean`
- `Leverage/Foundations.lean`
- `Leverage/Probability.lean`
- `Leverage/Theorems.lean`
- `Leverage/WeightedLeverage.lean`

Assumption bundles and fields.

- (No assumption bundles parsed.)

Conditional closure handles.

- (No conditional theorem handles parsed.)

First-principles Bayes derivation handles.

- (No Bayes-derivation theorem handles found.)

Compact Bayes handle IDs.

- (No compact Bayes alias IDs found.)